Chapter 8, Lecture 2

Marten Wegkamp

Method of Moments

Given $X_1, \ldots, X_n$ iid $f(x|\theta)$ with θ unknown.
In the last two lectures, we discussed properties (consistency, asymptotic distribution, unbiased estimators) of a given estimator. Now, we will be more systematic and discuss different types of estimation procedures.

For $k = 1, 2, \cdots$, we define μ_k as the kth moment of X and $\widehat{\mu}_k$ be its empirical counterpart. That is,

$$\begin{aligned} \mu_k &= \mathbb{E}[X^k] \\ \widehat{\mu}_k &= \frac{1}{n}\sum_{i=1}^{n} X_i^k \end{aligned}$$

Kolmogorov showed that, provided $\mathbb{E}[|X|^k] < \infty$ so that these quantities are well defined, and that the law of large numbers holds, that is,

$$\widehat{\mu}_k \xrightarrow{P} \mu_k, \qquad \text{as } n \to \infty$$

Now consider the scenario that the unknown parameter θ can be expressed in terms of some moments of X, that is,

$$\theta = \theta(\mu_1, \ldots, \mu_k)$$

for some k. The method of moments estimator simply replaces the unknown quantities μ_i with the known quantities $\widehat{\mu}_i$, computed from the data, and form consistent substitiues of μ_i. That is,

$$\widehat{\theta}_{MOM} = \theta(\widehat{\mu}_1, \ldots, \widehat{\mu}_k)$$

This is an example of a plug-in rule. The remainder of this lecture provides some examples of this rule.

EXAMPLE. If $X_1, \ldots, X_n$ are iid Exponential(λ), then

$$\lambda = 1/\mu_1$$

since $\mu_1 = \mathbb{E}(X) = 1/\lambda$, and

$$\widehat{\lambda}_{MOM} = 1/\widehat{\mu}_1 = 1/\overline{X}_n$$

as discussed before.

Notice that

$$\mu_2 = \mathbb{E}(X^2) = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = \frac{2}{\lambda^2}$$

and another choice could be the estimator

$$\widetilde{\lambda} = \sqrt{\frac{2}{\widehat{\mu}_2}} = \sqrt{\frac{2n}{\sum_{i=1}^n X_i^2}}$$

We prefer the first choice. Run some simulations in R and compare to MSEs to see for yourself. In general, we prefer to use the lower moment expression:
$t < -1000;\ l1 < -1 : t;\ l2 < -1 : t;$
for(i in $1 : t$){
$x < -\mathrm{rexp}(100, 2);$
$l1[i] < -1/\mathrm{mean}(x);$
$l2[i] < -\mathrm{sqrt}(2/\mathrm{mean}(x^2))\};$
$\mathrm{mse1} < -\mathrm{mean}((l1 - 2)^2);\ \mathrm{mse2} < -\mathrm{mean}(l2 - 2)^2);\ \mathrm{mse1};\ \mathrm{mse2};\ \mathrm{mse2/mse1}$

The *continuous mapping theorem* states:
Let $g(x)$ be a continuous function at $x = \mu$. If $T_n \xrightarrow{P} \mu$, then $g(T_n) \xrightarrow{P} g(\mu)$.
As we have seen in the previous lecture, this applies to our situation here: $\overline{X} \xrightarrow{P} \mu = 1/\lambda$ implies $\widehat{\lambda} = g(\overline{X}) \xrightarrow{P} \lambda$ since $g(x) = 1/x$ is continuous at $x = 1/\lambda \neq 0$.

EXAMPLE. If $X_1, \ldots, X_n$ are iid $N(\mu, \sigma^2)$. Since

$$\mu = \mu_1$$

we have

$$\widehat{\mu} = \widehat{\mu}_1 = \overline{X}_n.$$

Since

$$\sigma^2 = \mathrm{Var}(X) = \mu_2 - \mu_1^2,$$

we have

$$\begin{aligned}
\widehat{\sigma}^2_{MOM} &= \widehat{\mu}_2 - \widehat{\mu}_1^2 \\
&= \frac{1}{n}\sum_{i=1}^n X_i^2 - \left(\frac{1}{n}\sum_{j=1}^n X_j\right)^2 \\
&= \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X}_n)^2 \\
&= \frac{n-1}{n} S^2
\end{aligned}$$

where

$$S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X}_n)^2$$

is the sample variance.

Clearly, $\widehat{\mu} = \overline{X} \xrightarrow{P} \mu$ by the LLN. Use the continuous mapping theorem (for 2 variables) and the fact that $\widehat{\mu}_1 \xrightarrow{P} \mu_1$ and $\widehat{\mu}_2 \xrightarrow{P} \mu_2$ to show that

$$\widehat{\sigma}^2_{MOM} = \widehat{\mu}_2 - \widehat{\mu}_1^2 = g(\widehat{\mu}_1, \widehat{\mu}_2) \xrightarrow{P} g(\mu_1, \mu_2) = \sigma^2.$$

Since $S^2 = \widehat{\sigma}^2_{MOM} n/(n-1)$, we also have that S^2 is a consistent estimate of σ^2. In fact, we can show that it is unbiased.

EXAMPLE. Let $X_1, \ldots, X_n$ be iid Geometric(p). Recall

$$\mathbb{P}\{X = k\} = p(1-p)^{k-1}, \ k = 1, 2, \ldots$$

and $\mu_1 = \mathbb{E}[X] = 1/p$, so

$$p = 1/\mu_1$$

and

$$\widehat{p}_{MOM} = 1/\widehat{\mu}_1 = 1/\overline{X}_n$$

EXAMPLE. Let $X_1, \ldots, X_n$ be iid Gamma(α, λ). Recall $\mathbb{E}[X] = \alpha/\lambda$ and $\mathrm{Var}(X) = \alpha/\lambda^2$. This implies that

$$\mu_1 = \frac{\alpha}{\lambda} \qquad \text{and} \qquad \mu_2 - \mu_1^2 = \frac{\alpha}{\lambda^2}$$

This implies that

$$\frac{\mu_1}{\lambda} = \mu_2 - \mu_1^2 \Longrightarrow \lambda = \frac{\mu_1}{\mu_2 - \mu_1^2}$$

and

$$\alpha = \lambda\mu_1 = \frac{\mu_1^2}{\mu_2 - \mu_1^2}$$

Using the previous calculation

$$\widehat{\sigma}^2 := \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 = \widehat{\mu}_2 - \widehat{\mu}_1^2$$

we obtain

$$\begin{aligned} \widehat{\alpha}_{MOM} &= (\overline{X}_n)^2/\widehat{\sigma}^2 \\ \widehat{\lambda}_{MOM} &= \overline{X}_n/\widehat{\sigma}^2 \end{aligned}$$

We can argue that, using consistency of $\overline{X}_n \xrightarrow{P} \mu$ and $\widehat{\sigma}^2 \xrightarrow{P} \sigma^2$, and the continuous mapping theorem,

$$(\widehat{\alpha}_{MOM}, \widehat{\lambda}_{MOM}) = \Big(g_1(\overline{X}_n, \widehat{\sigma}^2), g_2(\overline{X}_n, \widehat{\sigma}^2)\Big) \xrightarrow{P} \Big(g_1(\mu, \sigma^2), g_2(\mu, \sigma^2)\Big) = (\alpha, \lambda).$$

Verify this for yourself. What are the functions g_1 and g_2?

EXAMPLE. Let $X_1, \ldots, X_n$ be iid Uniform$(0, \theta)$ with $\theta > 0$ unknown. Recall that the mean of X is the midpoint $\theta/2$ of the interval $(0, \theta)$. It is straightforward to see that

$$\widehat{\theta}_{MOM} = 2\widehat{\mu}_1 = 2\overline{X}_n$$

as $\theta = 2\mu_1$. For small samples, this may be problematic, and better estimates are possible.

For instance, consider $n = 3$, and $x_1 = 0.3$, $x_2 = 2.2$ and $x_3 = 0.5$. Then $\overline{X} = 1.0$ and $\widehat{\theta}_{MOM} = 2$.

Why do we know for a fact that the true parameter value must be *larger* than the estimate of 2?